Tirtha Nath

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OBJECTIVE

Computer Science Engineering student with a strong interest in AI and software development. I enjoy building applications that solve real-world problems and have worked on projects involving multi-agent systems, AI-powered applications, and full-stack development. Looking to apply my skills while continuing to grow as an engineer.

EDUCATION

SRM University — Kattankulathur, Tamil Nadu
B.Tech in Computer Science Engineering | *CGPA: 9.31* — 2023 – 2027

Accord School — Tirupati, Andhra Pradesh
Class XII: 75.2% | *Class X: 84.2%* — 2020 – 2023

EXPERIENCE

Wiingy Private Limited — Feb. 2026 – May 2026
Marketing Intern

- Spoke with users across online communities like Reddit and Whatsapp to answer questions, understand common issues, and gather feedback about the platform. **Certificate**
- Identified recurring user concerns and shared insights that helped improve community engagement and user experience.

Edu-versity — Feb. 2025 – April 2025
UI/UX Design Intern

- Designed user flows, wireframes, and interactive prototypes for five software projects using Figma. **Certificate**
- Worked with user requirements and feedback to design intuitive user interfaces that were easy to navigate, simple to understand, and focused on improving the overall user experience.

PROJECTS

Agent Marketplace | *FastAPI, React.js, AWS EC2, SQLite*

- Built a full-stack AI platform with six specialized agents communicating through Google's A2A protocol and Anthropic MCP.
- Developed secure JWT authentication, REST APIs, and a live dashboard to monitor agent communication and system activity.
- Deployed the application on AWS EC2 and built a scalable backend to support reliable communication between agents.

Disaster Management Multi-Agent System | *Multi-Agent Systems, Anthropic MCP, Google A2A*

- Built a collaborative network of four AI agents that communicate through the A2A protocol to coordinate disaster response.
- Developed a secure communication framework using Anthropic's Model Context Protocol (MCP) to connect AI agents with external data sources and support reliable communication. The framework was presented and published at IEEE ICICI 2026.
- Designed agent workflows that enabled information sharing and generated coordinated disaster response plans.

RouteXpert - An Explainable Symbolic AI Traffic Management System | *Python, Streamlit, NetworkX, Matplotlib, Pandas*

- Developed an explainable traffic management system for adaptive signal control using weighted priority scoring.
- Built an interactive dashboard to visualize congestion trends, routing decisions, and emergency vehicle prioritization.
- Implemented dynamic shortest-path routing to improve emergency vehicle movement across traffic networks. Results from this work were published at IEEE ICNWC 2026.

PUBLICATIONS & CERTIFICATIONS

- **Publication 1:** A. Jayakrishnan, T. Nath and B. M, "RouteXpert: An Explainable Symbolic AI Framework for Multi-Intersection Adaptive Traffic Control," *Third International Conference on Networking and Communications (ICNWC)*, 2026, pp. 15. DOI: 10.1109/ICNWC68145.2026.11518196. **Link**
- **Publication 2:** "Nath, T., Yadav, D., Damera, V., Balamurugan, G., & Jha, P. Integrated Framework for Secure Communication among Multiple Agents using MCP and A2A. In 2026 4th International Conference on Inventive Computing and Informatics (ICICI), IEEE, pp. 335342, June 2026. **Link**
- **Certifications: VLSI Physical Design (IIT Madras & SRM University)** | **SAP Generative AI Developer** | **Oracle Cloud Applications ERP Essentials** | **GitHub Copilot**.
- **Academic Courses:** NPTEL credentials in **Database Systems**, **Machine Learning**, and **Java Foundations**.

SKILLS

Programming Languages: Python, C++, C, SQL (MySQL, SQLite)
AI & Intelligent Systems: Large Language Models (LLMs), Multi-Agent Systems, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Anthropic MCP, Google A2A, Machine Learning, Symbolic AI
Frameworks & Technologies: FastAPI, Flask, React.js, Streamlit, REST APIs
Databases & Tools: MySQL, MongoDB, Supabase, Git, GitHub, AWS EC2, Figma
Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS)
Professional Skills: Problem Solving, Technical Communication, Teamwork, Adaptability